

Kapitsa pendulum — Solution

Y20 (2020) — English translation

A1

0.50 pts

The system has one degree of freedom, angle φ , and the position of the pendulum can be described by a function of this angle. Express the body's Cartesian coordinates $x(t)$ and $y(t)$ as functions of angle $\varphi(t)$ and time.

Answer: \textbf{Answer:}

$$\begin{aligned}x(t) &= l \sin \varphi \\y(t) &= a \cos \omega t - l \cos \varphi\end{aligned}$$

A2

0.50 pts

Find the velocity components v_x and v_y of the body. Express your answer in terms of angular velocity $\frac{d\varphi}{dt}$ and time.

Answer: \textbf{Answer:}

$$\begin{aligned}v_x &= \frac{dx}{dt} = l \cos \varphi \frac{d\varphi}{dt} \\v_y &= \frac{dy}{dt} = l \sin \varphi \frac{d\varphi}{dt} - \omega a \sin \omega t\end{aligned}$$

A3

0.50 pts

Find the kinetic and energy of the body. Express your answer in terms of angle φ , angular velocity $\frac{d\varphi}{dt}$ and time.

$$E_k = \frac{m}{2} \left(\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right)$$

Answer: \textbf{Answer:}

$$E_k = \frac{m}{2} \left(\left(l \frac{d\varphi}{dt} \right)^2 + \left(\omega a \sin \omega t \right)^2 - 2\omega a l \sin \omega t \sin \varphi \frac{d\varphi}{dt} \right)$$

A4

1.00 pts

Derive an expression for the instantaneous power N force applied to the pendulum's suspension point. Express your answer in terms of $m, g, l, a, \omega, \varphi, \frac{d\varphi}{dt}, \frac{d^2\varphi}{dt^2}$ and time.

Let us denote the suspension point of the pendulum as o . Then the expression for power becomes

$$N = \vec{F}_o \vec{v}_o = F_{oy} \frac{dy_o}{dt}$$

From the law изменения momentuma for тела

$$F_{oy} = mg + m \frac{d^2 y}{dt^2}$$

$$\frac{dy_o}{dt} = -\omega a \sin \omega t$$

$$\frac{d^2 y}{dt^2} = -\omega^2 a \cos \omega t + l \left(\sin \varphi \frac{d^2 \varphi}{dt^2} + \cos \varphi \left(\frac{d\varphi}{dt} \right)^2 \right)$$

Final answer

Answer: \textbf{Answer:}

$$N = -m\omega a \sin \omega t \left(g - \omega^2 a \cos \omega t + l \left(\sin \varphi \frac{d^2 \varphi}{dt^2} + \cos \varphi \left(\frac{d\varphi}{dt} \right)^2 \right) \right)$$

A5

1.00 pts

Show that the equation of motion is:

$$\frac{d^2 \varphi}{dt^2} = -\frac{1}{l}(g - a \omega^2 \cos \omega t) \sin \varphi.$$

The simplest way to show the validity of this relationship is associated with the transition to a non-inertial reference system associated with the suspension point. In it, in addition to the reaction force of the rod and the force of gravity, a vertical inertial force equal to $F_y = -m \frac{d^2 y_o}{dt^2} = m\omega^2 a \cos \omega t$ acts on the body.

From the law изменения момента импульса относительно точки подвеса we have

$$ml^2 \frac{d^2 \varphi}{dt^2} = (F_y - mg)l \sin \varphi$$

From записанных уравнений получается требуемое соотношение

$$\frac{d^2 \varphi}{dt^2} = -\frac{1}{l}(g - a \omega^2 \cos \omega t) \sin \varphi.$$

Answer and to the question can be more formally, differentiated the law of change of mechanical energy of the body in a stationary system of coordinates

$$N = \frac{dE}{dt} = mgl \sin \varphi \frac{d\varphi}{dt} + m \frac{dy}{dt} \frac{d^2 y}{dt^2} + m \frac{dx}{dt} \frac{d^2 x}{dt^2}$$

$$\frac{d^2 x}{dt^2} = l \left(\cos \varphi \frac{d^2 \varphi}{dt^2} - \sin \varphi \left(\frac{d\varphi}{dt} \right)^2 \right)$$

After setting all the quantities obtained earlier, we obtain the required ratio.

B1

1.00 pts

Show that the expression for the fast component is $\beta = -\frac{a}{l} \sin \gamma \cos \omega t$ (for an arbitrary value γ) is an approximate solution to the equation of motion from A5 under the above conditions. Consider an oscillation mode in which g is negligible compared to $a\omega^2$, i.e. $a\omega^2 \gg g$.

Since β is the fast component of

$$\frac{d^2 \beta}{dt^2} = \frac{d^2 \varphi}{dt^2}$$

$$\frac{d^2 \beta}{dt^2} = -\frac{1}{l}(g - a \omega^2 \cos \omega t) \sin(\gamma + \beta)$$

Since $\beta \ll \gamma$ and $g \ll a\omega^2$, equation for β takes the following form:

$$\frac{d^2 \beta}{dt^2} = \frac{\omega^2 a}{l} \sin \gamma \cos \omega t = -\omega^2 A \cos(\omega t)$$

where

$$A = -\frac{a}{l} \sin \gamma$$

From этого выражения for второй производной $\frac{d^2\beta}{dt^2}$ интегрированием we get expression for β :

$$\beta(t) = A \cos(\omega t) = -\frac{a}{l} \sin \gamma \cos \omega t$$

B2

2.50 pts

Combining the equation of motion from Item A5 s expansion of $\varphi = \gamma + \beta$ and with the expression from the previous paragraph, find an expression for $\frac{d^2\gamma}{dt^2}$ that describes the dynamics of the slow component γ averaging over a large number of fast oscillations with frequency ω , neglecting only terms of order a^n , where $n > 2$. Express $\frac{d^2\gamma}{dt^2}$ in terms of g, a, ω, l and γ . Simplify the final expression by counting $l\omega^2 \gg a\omega^2 \gg g$.

To average over a large number of fast oscillations

$$\frac{d^2\varphi}{dt^2} = \frac{d^2\gamma}{dt^2}$$

$$\frac{d^2\gamma}{dt^2} = -\frac{1}{lt} \int_0^t (g - a\omega^2 \cos \omega t) \sin \varphi (\sin \gamma \cos \beta + \cos \gamma \sin \beta) dt$$

For пренебрежения всеми членами порядка выше a^2 достаточно следующих сложений

$$\sin \beta = \beta \quad \cos \beta = 1 - \frac{\beta^2}{2}$$

Expression после сложений and раскрытия скобок for/atнимает вид

$$\frac{d^2\gamma}{dt^2} = -\frac{1}{lt} \int_0^t \left(g \sin \gamma \left(1 - \frac{a^2 \sin^2 \gamma}{l^2} \cos^2 \omega t \right) + a\omega^2 \left(\frac{a^2 \sin^2 \gamma}{2l^2} \cos^3 \omega t + \frac{a \sin 2\gamma}{2l} \cos^2 \omega t \right) \right) dt$$

For/At интегрировании учтём малость $\frac{a^2}{l^2}$, а also что for/at большом времени усреднения средние значения тригонометрических функций в нечётных степенях equal нулю, это позволяет ссу отбросить большинство слагаемых.

$$\frac{d^2\gamma}{dt^2} = -\frac{1}{lt} \int_0^t \left(g \sin \gamma + \frac{a^2 \omega^2 \sin 2\gamma}{4l} (1 + \cos 2\omega t) \right) dt$$

We get the answer

Answer: \textbf{Answer:}

$$\frac{d^2\gamma}{dt^2} = -\frac{1}{l} \left(g \sin \gamma + \frac{a^2 \omega^2 \sin 2\gamma}{4l} \right)$$

B3

0.70 pts

Obtain an expression for the time-averaged moment of all forces M_O acting on the pendulum relative to the suspension point. Express your answer in terms of m, g, l, a, ω and γ .

Returning to point A5

$$ml^2 \frac{d^2\varphi}{dt^2} = M$$